

CHADRON DUZI

ENGINEERING DESIGN PORTFOLIO

Mechanical Engineering | UC Irvine | Wearable Devices | Product Design | Rapid Prototyping

ABOUT

Third year Mechanical Engineering student at UC Irvine focused on turning early concepts into testable physical systems. My work spans wearable sensing, CAD, 3D printing, electromechanical prototyping, and iterative validation. I am especially interested in medical and rehabilitation technology where mechanical design and sensing must work together reliably on the body.

SELECTED WORK

01	Dorsal Force-Sensing Hand Exoskeleton Wearable sensing system designed to estimate PIP-joint flexion from dorsal knuckle loading while preserving the palmar surface. SolidWorks • 3D Printing • Electronics • ImageJ • Excel
02	Hand Tremor Mitigation Device SolidWorks assembly centered on a tuned mass damper concept with various mechanical components. SolidWorks • Assembly Design

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01 DORSAL FORCE-SENSING HAND EXOSKELETON

Independent Research Project; Presenting work at IMECE 2026 In Vancouver, BC, Canada in November 2026.

PROBLEM

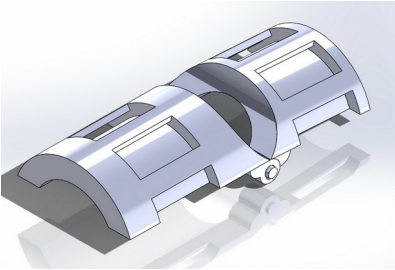
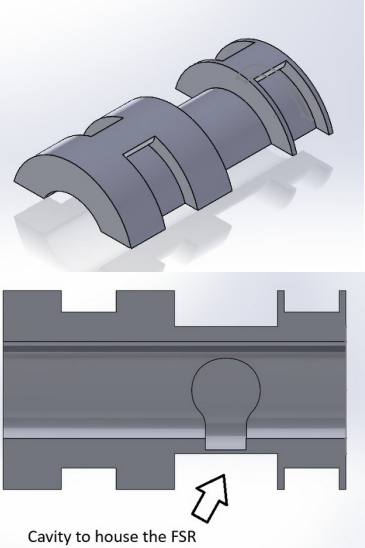
Assistive hand devices are commonly used by individuals with impaired hand function due to injury, neurological conditions, or reduced motor control, and often work by detecting a user’s intended movement and then providing mechanical assistance. Many existing devices rely on sensors attached to the palm or fingertips to predict hand motion, but these signals may become useful only after object contact and can interfere with natural tactile interaction. The design question was whether a dorsally attached sensor could provide an earlier indication of intended movement without covering the palm or requiring the user to first contact an object.

SOLUTION

A proof-of-concept index-finger exoskeleton positioned a force-sensing resistor (FSR) over the dorsal proximal interphalangeal (PIP) joint. The exoskeleton was iteratively redesigned to improve sensor support, attachment repeatability, and loading consistency so FSR output could be mapped to finger flexion angle. The finger flexion angle could theoretically be used to determine when and how the user is moving their hand, which could signal to an actuator to provide assistance. ‘

DESIGN EVOLUTION

Provided below is a table detailing the 4 primary versions of the hand exoskeleton.

V1 - Polylactic Acid (PLA) Model with pin joint		V2 - Thermoplastic Polyurethane (TPU) Model with curved backing	
Physical Prototype	CAD Model	Physical Prototype	CAD Model
			
Design Change + Rationale: The initial prototype was used to establish the overall geometry and basic fit on the dorsal side of the hand. PLA was selected as it was the only material available at the time.	Result: The rigid structure helped validate the general concept, but it did not have a structure to house the FSR. This motivated the transition to a flexible material.	Design Change + Rationale: Replaced PLA with TPU and introduced a curved backing intended to match the natural contour of the hand and provide a housing for the FSR via a cavity on the side contacting the hand.	Result: TPU improved comfort and removed the need for a pin joint (decreasing complexity), but the curved backing produced local buckling and unwanted creasing, creating excessive bending of the FSR. The FSR also could not be reliably secured by the cavity alone, and knuckle contact created a concentrated load rather than distributing force across

the sensor. These issues motivated a flat backing, improved FSR retention, and an intermediate material between the knuckle and FSR

V3 - TPU Model with flat backing

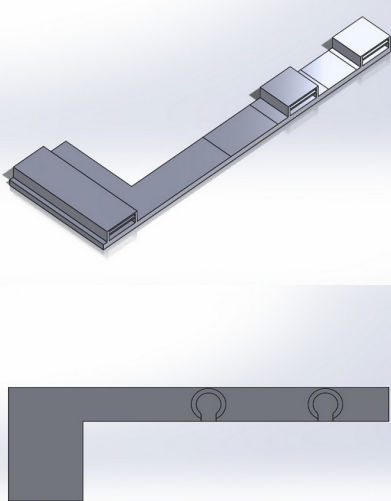
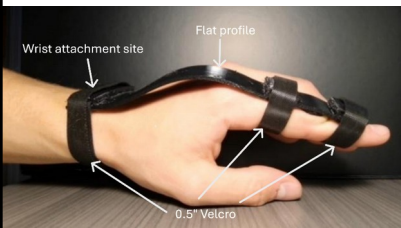
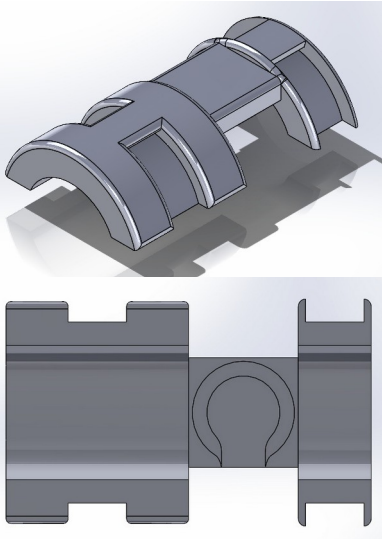
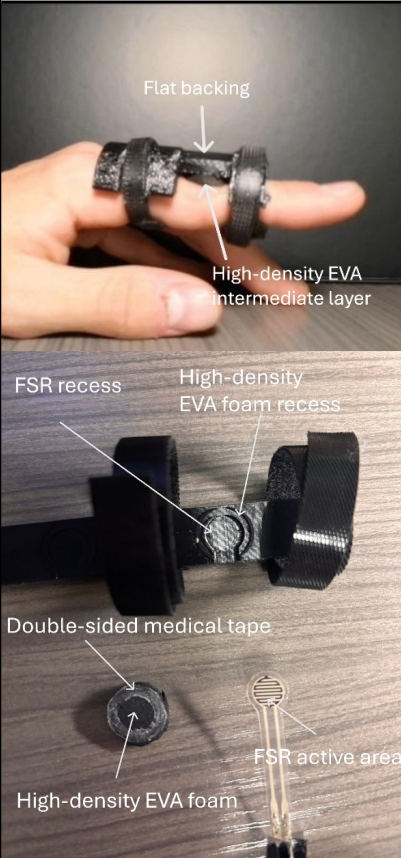
V4 - TPU Model with wrist attachment site

Physical Prototype

CAD Model

Physical Prototype

CAD Model



Design Change + Rationale: The backing was changed from curved to flat so the TPU could conform naturally to the hand during deformation, reducing the buckling observed in the previous design. An intermediate ethylene-vinyl acetate (EVA) layer was added to distribute knuckle force over a larger area of the FSR. EVA was selected because of its low cost, availability, and relatively low compression set. A lower compression set reduces permanent deformation after repeated loading, helping preserve the geometry of the interface and improve measurement repeatability across multiple trials using

Result: The flat TPU backing reduced the local buckling observed in V2, allowing the structure to conform more naturally to the dorsal surface of the hand during finger motion and reduce unintended bending of the FSR. The addition of the intermediate EVA layer distributed the knuckle force over a larger portion of the sensor, creating a more uniform loading condition and reducing stress concentration at the point of contact. The revised retention method, using a ring of double-sided medical tape around the perimeter of the FSR, also improved sensor stability while avoiding adhesive contact

Design Change + Rationale: A wrist attachment was added to better anchor the exoskeleton and reduce the axial sliding observed in V3 during repeated finger flexion and extension. Wider velcro straps were also introduced to increase the contact area and friction between the device and the user's hand, further resisting unwanted longitudinal movement and helping maintain more consistent alignment between the knuckle and FSR.

Result: The wrist attachment and wider Velcro improved overall device stability and reduced axial sliding during repeated motion, helping maintain more consistent sensor positioning. However, the increased strap width introduced a tradeoff between stability and wearability. Compared with the narrower 0.25-inch Velcro used previously, the wider straps were less comfortable and restricted finger mobility to a greater extent. This highlighted the need to balance secure attachment with user comfort and range of motion in future design iterations.

the same exoskeleton. FSR retention was also improved by introducing a layer of double-sided medical tape between the EVA foam, FSR, and exoskeleton. The tape was cut in a circular ring around the perimeter of the FSR, leaving the active sensing area exposed. This prevented the adhesive from directly contacting the active region of the sensor, which could otherwise introduce unwanted tensile loading or constrain sensor deformation and reduce signal reliability.

with the active area. Together, these changes produced a more stable and repeatable sensing interface. However, the overall exoskeleton still experienced axial sliding along the hand after repeated cycles of finger flexion and extension, causing gradual changes in device position relative to the knuckle. This remaining issue motivated the addition of a wrist attachment in V4 to better anchor the device and maintain consistent sensor alignment during repeated motion.

DATA ACQUISITION SETUP

Voltage Divider & ADC Output

A FSR changes resistance as force is applied: greater force produces lower resistance. In this project, the FSR was positioned on the dorsal side of the index finger's PIP joint. As the user flexes their index finger, the knuckle protrusion increases, increasing the force on the FSR and decreasing the resistance. To convert this resistance change into a measurable signal, a simple voltage-divider circuit was used, as shown in the figure to the

left. The output voltage is defined as $V_{output} = V_{input} \frac{R_i}{R_i + R_{FSR}}$.

With a 3.3 V input, the output voltage varies between approximately 0 and 3.3 V depending on the FSR resistance. Since an ESP32 was used as the microcontroller, its 12-bit analog-to-digital converter (ADC) converts this voltage into a digital reading from 0–4095 counts. These readings were recorded at different finger joint angles so the sensor response could later be compared with joint position and used to develop a relationship

between flexion angle and ADC output.

FSR Selection

Two force-sensing resistors were initially considered: the DF9-40 (right), specified for approximately 0–500 g, and the RP-C5-LT-LF (left), specified for approximately 5–600 g. Both force ranges were suitable for the relatively small loads expected from dorsal knuckle protrusion during finger flexion, so sensing geometry became the more important selection factor. The RP-C5 had a 5 mm active sensing area, while the DF9-40 provided a larger 7.5 mm diameter active area. Because the sensor needed to capture loading across the protruding proximal interphalangeal (PIP) joint, the larger sensing area of the DF9-40 provided better

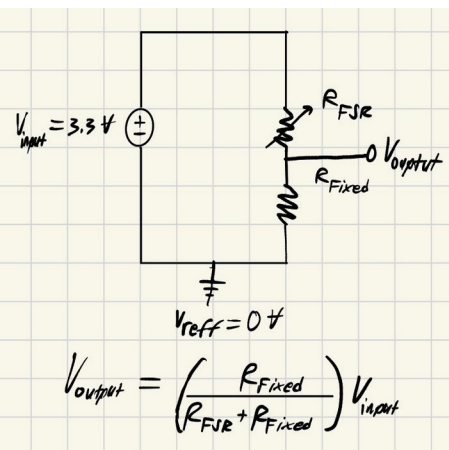
coverage and was therefore selected.

Fixed Resistor Selection

Manufacturer data for the DF9-40 was used to estimate its resistance across the expected loading range. A 22 kΩ fixed resistor was selected so the voltage-divider output would use a large portion of the ESP32's 0–4095 analog-to-digital converter (ADC) range without reaching the upper measurement limit. If the signal reaches that limit, the ADC becomes saturated, meaning further increases in voltage can no longer be distinguished. This results in clipping, where the recorded signal is effectively capped at the maximum ADC value. Based on the sensor's nominal 0–500 g force range, the circuit was predicted to produce approximately 0–3500 ADC counts. During testing, the actual knuckle-loading range was closer to 0–300 g, which produced approximately 0–3300 counts. This confirmed that the 22 kΩ resistor avoided clipping while still using much of the available ADC range for useful sensitivity.

Experimental Setup

The exoskeleton was secured to the hand with the force-sensing resistor (FSR) centered over the dorsal proximal

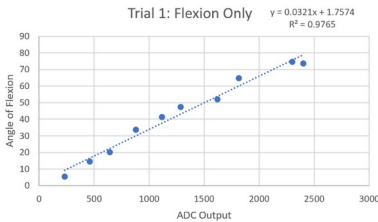
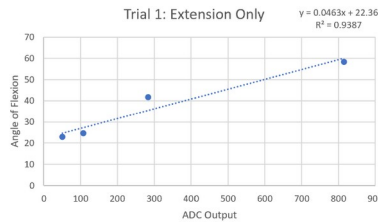
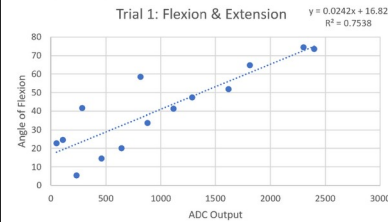


interphalangeal (PIP) joint. Design V3 was used for Trial 1, while Design V4 was used for Trials 2, 3, and 4 after Trial 1 revealed approximately 5 mm of axial sliding. During repeated finger flexion and extension, a webcam recorded the index finger while the ESP32 simultaneously streamed analog-to-digital converter (ADC) readings through the Arduino IDE. Both were captured together in OBS Studio so each sensor reading could later be matched to the corresponding finger position. A button was used to instantaneously record the live ADC value at selected finger positions, making it easier to document discrete measurements throughout the motion cycle.

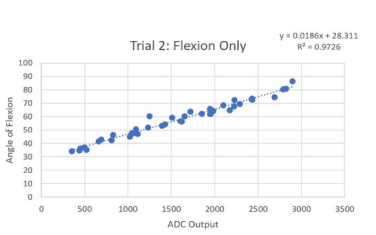
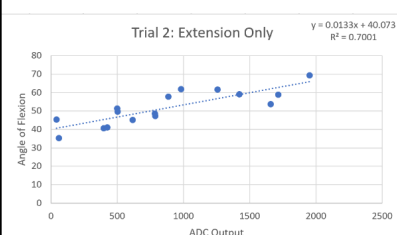
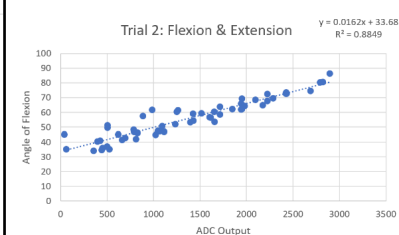
Joint Angle Measurement

The recorded video was reviewed frame-by-frame, and screenshots were captured at each selected sensor reading. ImageJ was then used to measure the PIP joint angle using three anatomical landmarks positioned near the metacarpophalangeal (MCP) joint, PIP joint, and distal interphalangeal (DIP) joint. Two measurements were taken for each image in opposite directions and averaged to reduce measurement bias. The measured angles and corresponding ADC readings were then organized in Excel for regression and prediction analysis.

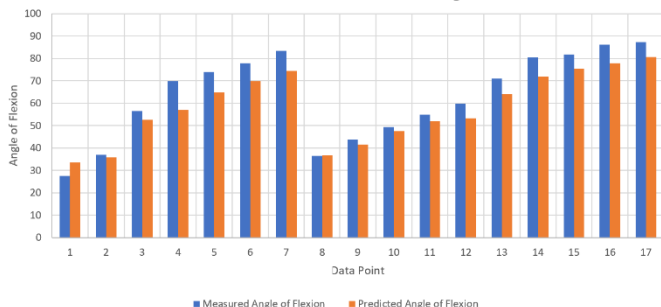
VALIDATION & RESULTS

Trial 1	Flexion Only	Extension Only	Flexion + Extension
Purpose: Determine if ADC Output changed predictably with PIP flexion.			
1 Flexion & Extension Cycle	9 Data Points	4 Data Points	13 Data Points
Graphs of Angle of Flexion vs ADC Output Accompanied by a linear relationship and R ² values in the top right corner.			
	R ² Values	0.9765	0.9387
Takeaways: Trial 1 demonstrated a strong relationship between joint angle and FSR output when flexion and extension were analyzed independently. However, approximately 5 mm of axial sliding occurred after the first cycle, limiting repeatability and motivating the more stable V4 attachment design used in subsequent trials.			

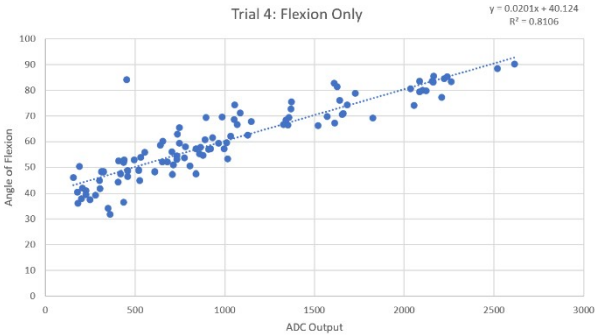
Trial 2	Flexion Only	Extension Only	Flexion + Extension
Purpose: Purpose: Evaluate whether the improved V4 attachment design produced a repeatable relationship between PIP joint angle and ADC output across multiple cycles.			
6 Flexion & Extension Cycles	41 Data Points	16 Data Points	57 Data Points

Graphs of Angle of Flexion vs ADC Output Accompanied by a linear relationship and R ² values in the top right corner.			
R² Values	0.9726	0.7001	0.8849

Takeaways: Trial 2 showed that the V4 setup produced a strong and repeatable relationship during flexion across multiple motion cycles. Extension was substantially less consistent, suggesting that unloading of the sensor was more sensitive to small changes in device position or interface conditions. Because flexion produced the strongest relationship, the flexion-only model was carried forward for prediction testing in Trial 3.

Trial 3	(Measured) Independent Flexion Only Dataset		(Predicted) Trial 2 Flexion Only Dataset	
2 Flexion Trials	20 Data Points		Applies $y = 0.0186x + 28.31$, where x is the ADC Output and y is the predicted Angle of Flexion	
Bar Graph of Measured Angle of Flexion vs Predicted Angle of Flexion		Trial 3: Measured vs Predicted Angle of Flexion 		
Results	$R^2 = 0.9776$	Mean Absolute Error (MAE) = 5.89°	Root Mean Square Error (RMSE) = 6.75°	Mean Bias = -5.16°
Purpose: Determine whether the flexion relationship developed in Trial 2 could predict joint angle from ADC output in a new dataset.				
Takeaways: Trial 3 produced an R^2 of 0.9776, showing that the predicted and measured flexion angles followed a very strong relationship. The model achieved a MAE of 5.89° and a RSME of 6.75° , indicating that most predictions were within several degrees of the measured joint angle. However, the mean bias of -5.16° showed that the error was not purely random—the model consistently tended to underpredict flexion angle. Overall, the results demonstrated that dorsal FSR output could estimate joint angle with promising consistency, while also revealing that improved control of device placement, strap tension, and sensor loading would be needed to reduce systematic error.				
Trial 4		Flexion Only		
20 Flexion Trials		100 Data Points		

Graph of
Angle of Flexion
vs
ADC Output



R² Value

0.8106

Purpose: Determine whether increasing the number of trials and total data points would strengthen the relationship between ADC output and PIP flexion angle.

Takeaways: Despite increasing the dataset to 100 flexion data points, Trial 4 produced an R² of 0.8106, which was lower than the 0.9726 obtained during Trial 2. This showed that collecting more data did not automatically improve the strength of the relationship between ADC output and joint angle. Instead, the reduced fit suggested that trial-to-trial variability such as changes in device placement, strap tension, sensor alignment, or deformation of the EVA interface had a larger effect on model consistency than dataset size alone.

Overall Project Takeaways

The project demonstrated that a dorsally mounted force-sensing resistor (FSR) can provide a measurable relationship with proximal interphalangeal (PIP) joint flexion, supporting its use as a potential method to predict hand movement in assistive devices. The strongest results occurred during flexion, with Trial 2 producing an R² of 0.9726 and the independent prediction test in Trial 3 achieving a mean absolute error (MAE) of 5.89°. The testing also showed that mechanical consistency was more important than simply collecting more data. Trial 4 used 100 flexion data points but produced a lower R² of 0.8106, indicating that small changes in device placement, strap tension, sensor alignment, and interface deformation could significantly affect the sensor response. The main takeaway was that the sensing concept itself was feasible, but reliable performance depended heavily on the mechanical design of the wearable interface. Future work would focus on improving attachment repeatability, reducing sensor-interface variability, and validating the system across additional users, fingers, and joints

Key Limitations & Next Steps

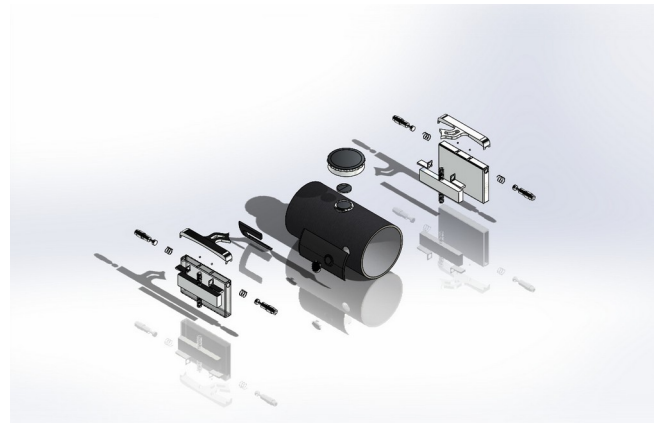
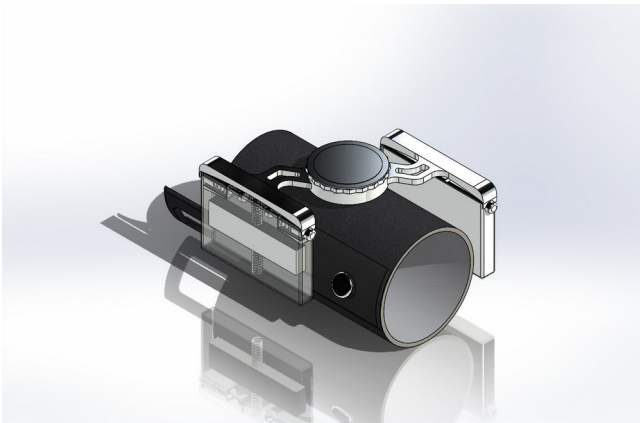
The current study was limited to a single user and a single index-finger PIP joint, so the results cannot yet be generalized across different users, finger sizes, or joints. Joint angle was also measured manually from video using ImageJ, which introduced potential error from landmark placement and frame selection. In addition, exoskeleton fit and strap tension were adjusted manually, and repeated loading may have changed the EVA interface and FSR response over time. Finally, the entire project was bootstrapped, highly limiting what materials were accessible.

02 HAND TREMOR MITIGATION DEVICE

Engineering Graphics Final Project

Project Overview

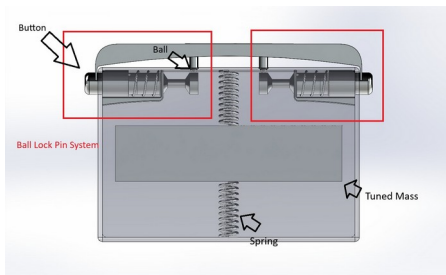
This project focused on developing a compact, wrist worn mechanical device intended to reduce hand tremor using tuned mass dampers (TMD). The system was designed in SolidWorks, incorporating spring mass damper systems, ball lock pin systems, a detachable bayonet mounted central assembly, and a wrist sleeve.



DESIGN OVERVIEW

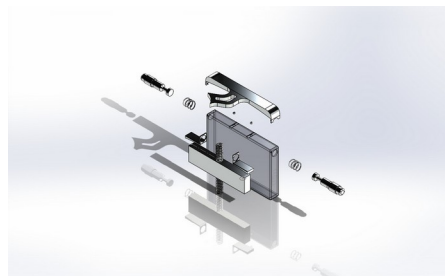
Tuned Mass Damper System

The core of the device is a TMD subassembly consisting of paired moving masses and springs constrained within the housing. When the device experiences oscillatory motion, the TMD systems are designed to oscillate relative to the main structure. If the natural frequency of the mass is tuned near the frequency of the unwanted tremor, the moving mass develops an opposing dynamic response that reduces the vibration transmitted through the primary structure.



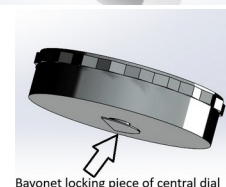
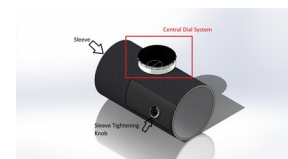
Ball Lock Pin System

The ball lock pin systems in this device allow it to be collapsible for storage purposes. When the buttons on either side are pressed, the pins move towards the midline, allowing the balls to drop into the cavity. The balls now no longer provide rotational resistance and the device can now rotate up to collapse.



Sleeve & Central Dial System

The sleeve allowed the user to wear the device on their wrist. The central dial allows the mechanical system to be detached from the sleeve for storage purposes. The central dial is attached to the TMD systems, and is connected via a bayonet lock. The tightening knob would theoretically have elastic cables connected to it, allowing the sleeve to be tightened, although the cables were not modeled in SolidWorks.



Project Outcome & Limitations

The final SolidWorks model integrated the TMD system, collapsible ball-lock mechanism, detachable bayonet mounted central assembly, structural shell, and wrist sleeve into a single compact wearable concept. The project was a theoretical design of a wrist worn device to mitigate tremors. The design was not physically fabricated or experimentally validated. As a result, the project did not verify actual tremor reduction, tuned-frequency performance, comfort, or long-term mechanical reliability. Future work would require physical prototyping, tuning the mass-spring system to the target tremor frequency, and testing the device on users to evaluate both vibration reduction and wearability.